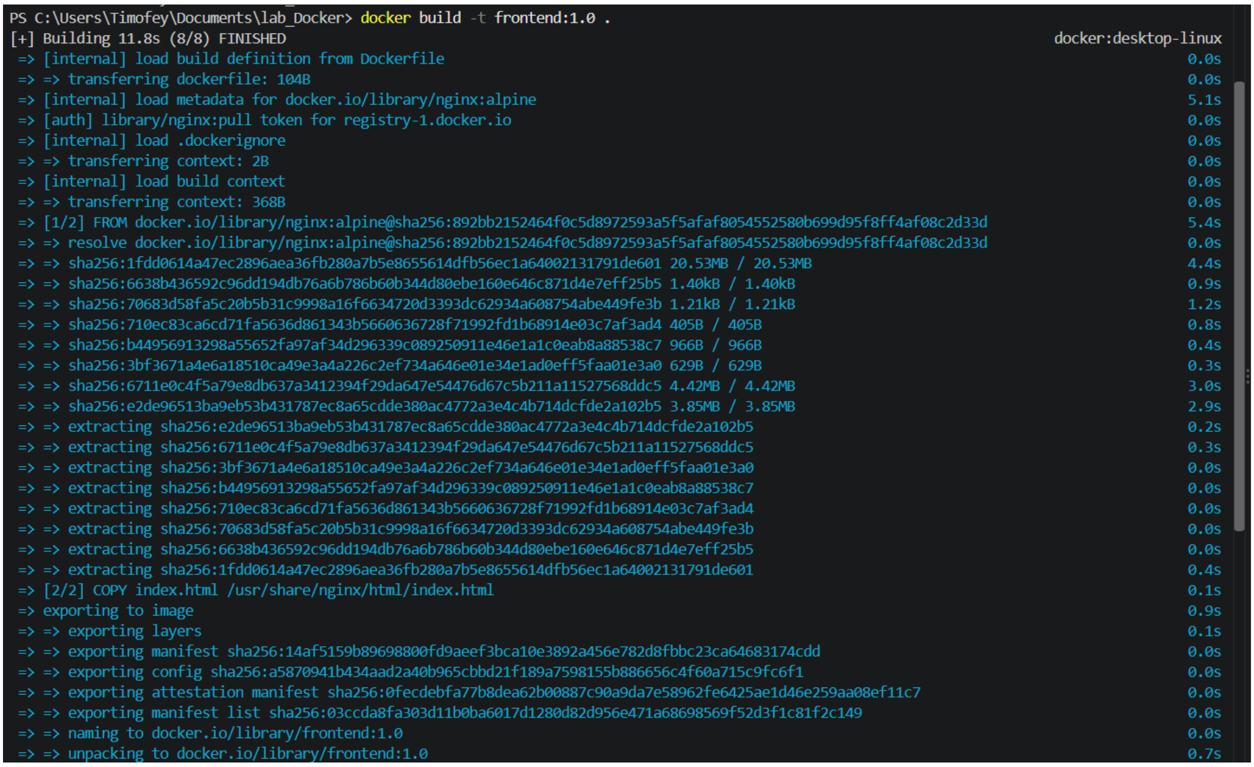
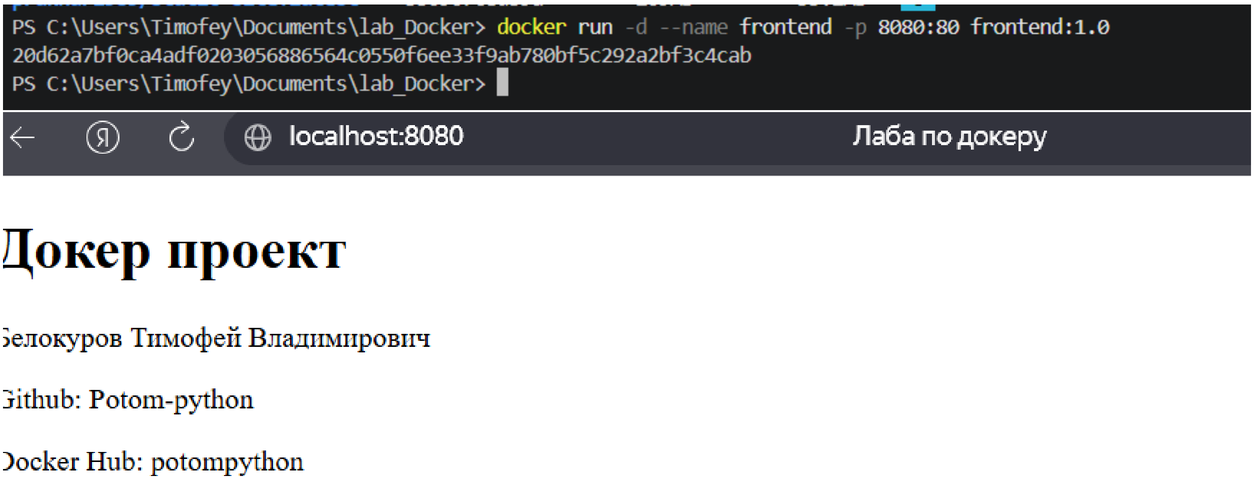
1.1



```
PS C:\Users\Timofey\Documents\lab_Docker> docker build -t frontend:1.0 .
[+] Building 11.8s (8/8) FINISHED                                                                                              docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                                           0.0s
 => => transferring dockerfile: 104B                                                                                                           0.0s
 => [internal] load metadata for docker.io/library/nginx:alpine                                                                                5.1s
 => [auth] library/nginx:pull token for registry-1.docker.io                                                                                   0.0s
 => [internal] load .dockerignore                                                                                                              0.0s
 => => transferring context: 2B                                                                                                                0.0s
 => [internal] load build context                                                                                                              0.0s
 => => transferring context: 368B                                                                                                              0.0s
 => [1/2] FROM docker.io/library/nginx:alpine@sha256:892bb2152464f0c5d8972593a5f5afaf8054552580b699d95f8ff4af08c2d33d                          5.4s
 => => resolve docker.io/library/nginx:alpine@sha256:892bb2152464f0c5d8972593a5f5afaf8054552580b699d95f8ff4af08c2d33d                          0.0s
 => => sha256:1fdd0614a47ec2896aea36fb280a7b5e8655614dfb56ec1a64002131791de601 20.53MB / 20.53MB                                               4.4s
 => => sha256:6638b436592c96dd194db76a6b786b60b344d80ebe160e646c871d4e7eff25b5 1.40kB / 1.40kB                                                 0.9s
 => => sha256:70683d58fa5c20b5b31c9998a16f6634720d3393dc62934a608754abe449fe3b 1.21kB / 1.21kB                                                 1.2s
 => => sha256:710ec83ca6cd71fa5636d861343b5660636728f71992fd1b68914e03c7af3ad4 405B / 405B                                                     0.8s
 => => sha256:b44956913298a55652fa97af34d296339c089250911e46e1a1c0eab8a88538c7 966B / 966B                                                     0.4s
 => => sha256:3bf3671a4e6a18510ca49e3a4a226c2ef734a646e01e34e1ad0eff5faa01e3a0 629B / 629B                                                     0.3s
 => => sha256:6711e0c4f5a79e8db637a3412394f29da647e54476d67c5b211a11527568ddc5 4.42MB / 4.42MB                                                 3.0s
 => => sha256:e2de96513ba9eb53b431787ec8a65cdde380ac4772a3e4c4b714dcfde2a102b5 3.85MB / 3.85MB                                                 2.9s
 => => extracting sha256:e2de96513ba9eb53b431787ec8a65cdde380ac4772a3e4c4b714dcfde2a102b5                                                      0.2s
 => => extracting sha256:6711e0c4f5a79e8db637a3412394f29da647e54476d67c5b211a11527568ddc5                                                      0.3s
 => => extracting sha256:3bf3671a4e6a18510ca49e3a4a226c2ef734a646e01e34e1ad0eff5faa01e3a0                                                      0.0s
 => => extracting sha256:b44956913298a55652fa97af34d296339c089250911e46e1a1c0eab8a88538c7                                                      0.0s
 => => extracting sha256:710ec83ca6cd71fa5636d861343b5660636728f71992fd1b68914e03c7af3ad4                                                      0.0s
 => => extracting sha256:70683d58fa5c20b5b31c9998a16f6634720d3393dc62934a608754abe449fe3b                                                      0.0s
 => => extracting sha256:6638b436592c96dd194db76a6b786b60b344d80ebe160e646c871d4e7eff25b5                                                      0.0s
 => => extracting sha256:1fdd0614a47ec2896aea36fb280a7b5e8655614dfb56ec1a64002131791de601                                                      0.4s
 => [2/2] COPY index.html /usr/share/nginx/html/index.html                                                                                     0.1s
 => exporting to image                                                                                                                         0.9s
 => => exporting layers                                                                                                                        0.1s
 => => exporting manifest sha256:14af5159b89698800fd9aeef3bca10e3892a456e782d8fbbc23ca64683174cdd                                              0.0s
 => => exporting config sha256:a5870941b434aad2a40b965cbbd21f189a7598155b886656c4f60a715c9fc6f1                                                0.0s
 => => exporting attestation manifest sha256:0fecdebfa77b8dea62b00887c90a9da7e58962fe6425ae1d46e259aa08ef11c7                                  0.0s
 => => exporting manifest list sha256:03ccda8fa303d11b0ba6017d1280d82d956e471a68698569f52d3f1c81f2c149                                         0.0s
 => => naming to docker.io/library/frontend:1.0                                                                                                0.0s
 => => unpacking to docker.io/library/frontend:1.0                                                                                             0.7s
```

1.2



```
PS C:\Users\Timofey\Documents\lab_Docker> docker run -d --name frontend -p 8080:80 frontend:1.0
20d62a7bf0ca4adf0203056886564c0550f6ee33f9ab780bf5c292a2bf3c4cab
PS C:\Users\Timofey\Documents\lab_Docker>
```

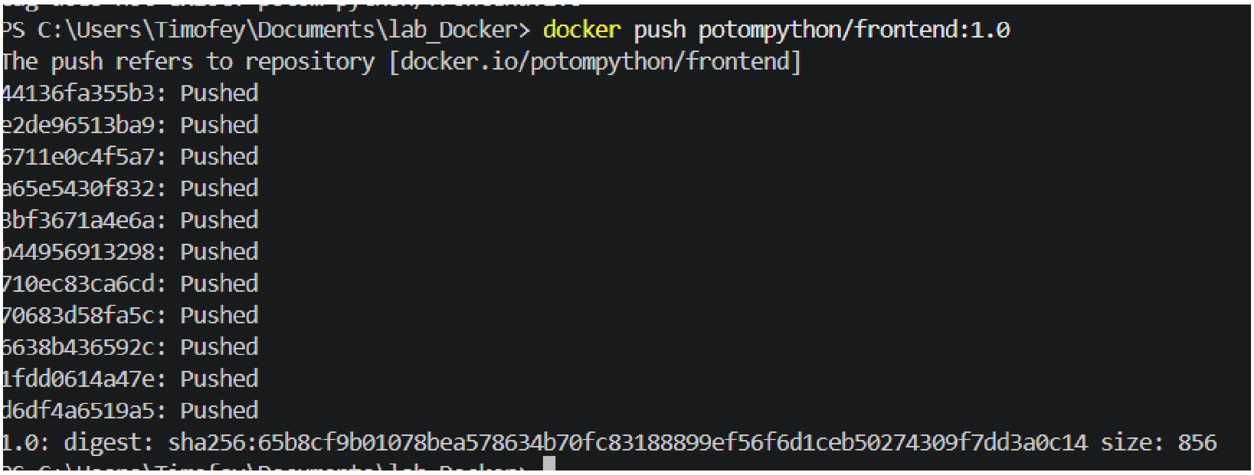
Докер проект

елокуров Тимофей Владимирович

ithub: Potom-python

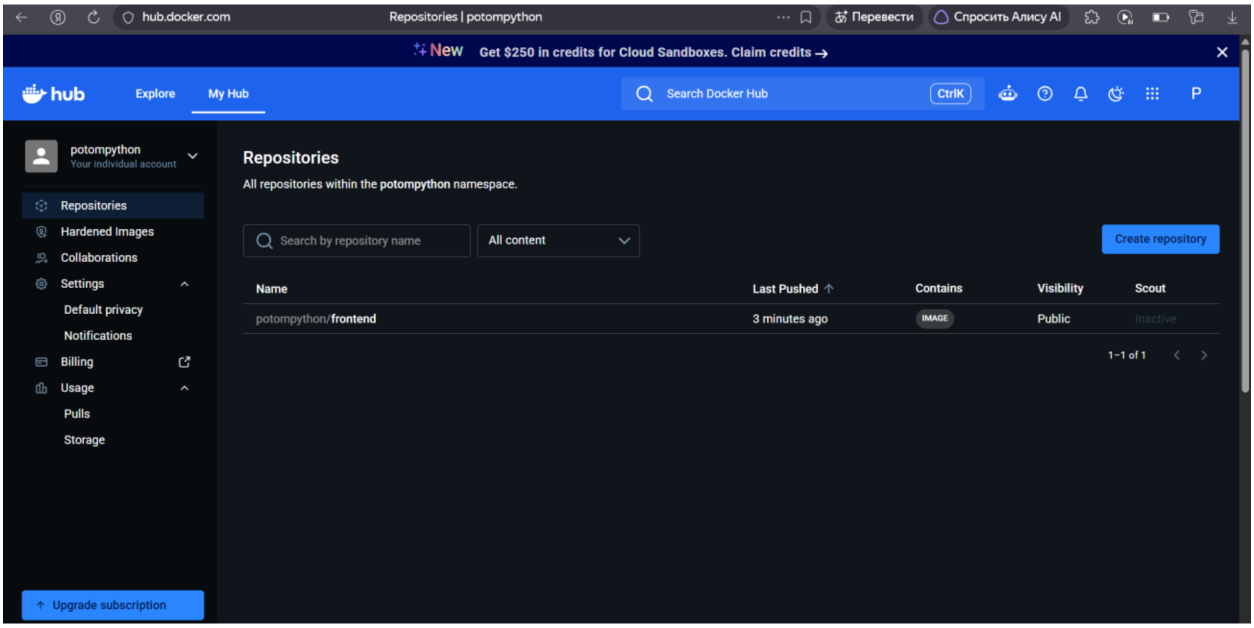
ocker Hub: potompython

1.3



```
PS C:\Users\Timofey\Documents\lab_Docker> docker push potompython/frontend:1.0
The push refers to repository [docker.io/potompython/frontend]
44136fa355b3: Pushed
e2de96513ba9: Pushed
6711e0c4f5a7: Pushed
a65e5430f832: Pushed
3bf3671a4e6a: Pushed
b44956913298: Pushed
710ec83ca6cd: Pushed
70683d58fa5c: Pushed
6638b436592c: Pushed
1fdd0614a47e: Pushed
d6df4a6519a5: Pushed
1.0: digest: sha256:65b8cf9b01078bea578634b70fc83188899ef56f6d1ceb50274309f7dd3a0c14 size: 856
PS C:\Users\Timofey\Documents\lab_Docker>
```

1.4



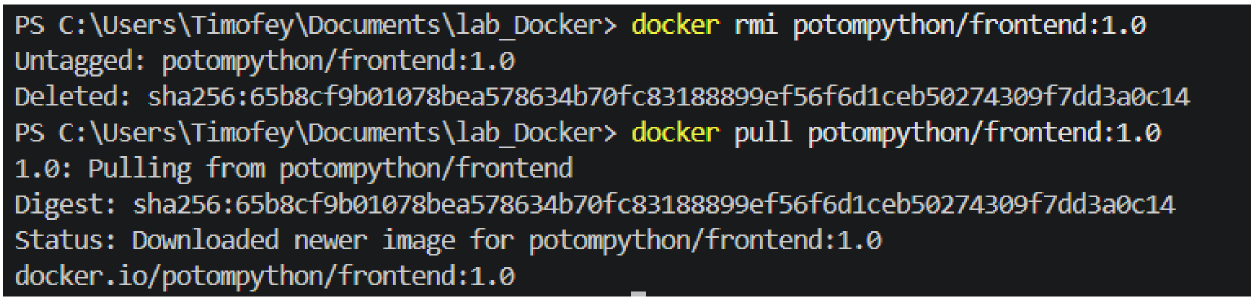
hub.docker.com
Repositories | potompython
Перевести
Спросить Алису AI
New Get $250 in credits for Cloud Sandboxes. Claim credits →
hub
Explore
My Hub
Search Docker Hub
CtrlK
P
potompython
Your individual account
Repositories
Hardened Images
Collaborations
Settings
Default privacy
Notifications
Billing
Usage
Pulls
Storage
Upgrade subscription
Search by repository name
All content
Create repository
1–1 of 1

Repositories

All repositories within the **potompython** namespace.

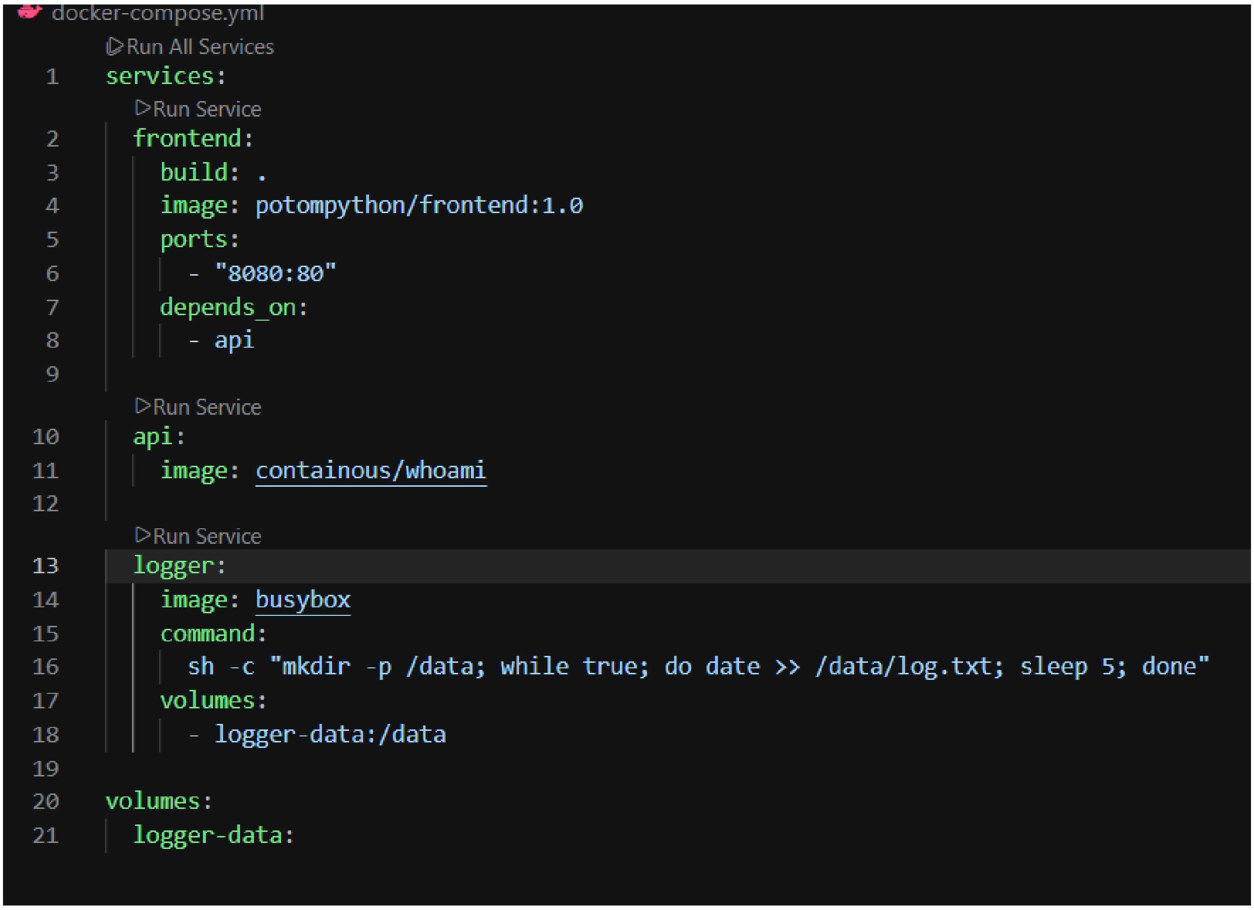
Name	Last Pushed ↑	Contains	Visibility	Scout
potompython/**frontend**	3 minutes ago	IMAGE	Public	Inactive

1.5



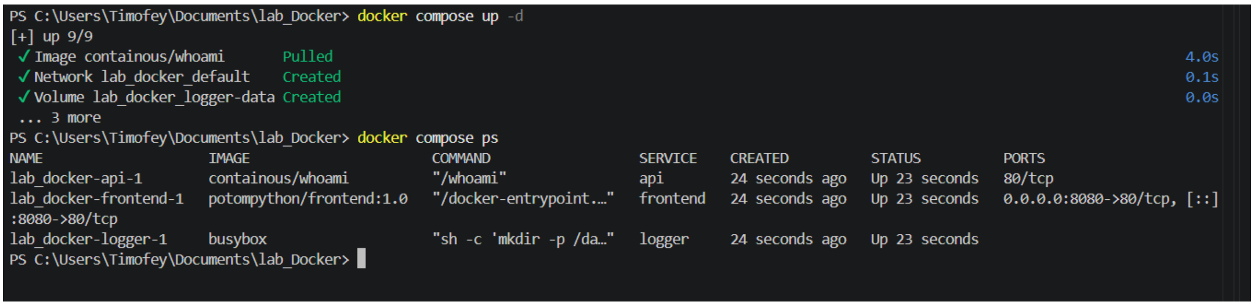
```
PS C:\Users\Timofey\Documents\lab_Docker> docker rmi potompython/frontend:1.0
Untagged: potompython/frontend:1.0
Deleted: sha256:65b8cf9b01078bea578634b70fc83188899ef56f6d1ceb50274309f7dd3a0c14
PS C:\Users\Timofey\Documents\lab_Docker> docker pull potompython/frontend:1.0
1.0: Pulling from potompython/frontend
Digest: sha256:65b8cf9b01078bea578634b70fc83188899ef56f6d1ceb50274309f7dd3a0c14
Status: Downloaded newer image for potompython/frontend:1.0
docker.io/potompython/frontend:1.0
```

2.1



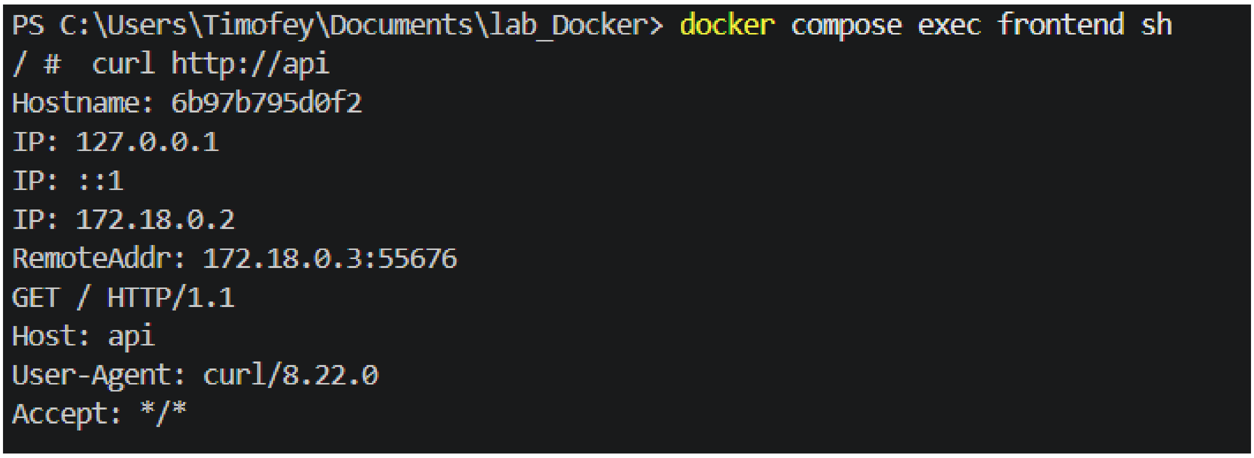
```
services:
  frontend:
    build: .
    image: potompython/frontend:1.0
    ports:
      - "8080:80"
    depends_on:
      - api

  api:
    image: containous/whoami

  logger:
    image: busybox
    command:
      sh -c "mkdir -p /data; while true; do date >> /data/log.txt; sleep 5; done"
    volumes:
      - logger-data:/data

volumes:
  logger-data:
```

2.2



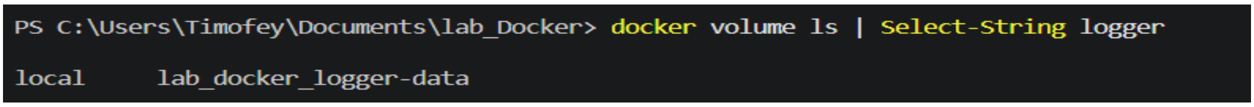
```
PS C:\Users\Timofey\Documents\lab_Docker> docker compose up -d
[+] up 9/9
 ✓ Image containous/whoami       Pulled                                                                  4.0s
 ✓ Network lab_docker_default    Created                                                                 0.1s
 ✓ Volume lab_docker_logger-data Created                                                                 0.0s
 ... 3 more
PS C:\Users\Timofey\Documents\lab_Docker> docker compose ps
NAME                    IMAGE                      COMMAND                   SERVICE    CREATED          STATUS          PORTS
lab_docker-api-1        containous/whoami          "/whoami"                 api        24 seconds ago   Up 23 seconds   80/tcp
lab_docker-frontend-1   potompython/frontend:1.0   "/docker-entrypoint.…"    frontend   24 seconds ago   Up 23 seconds   0.0.0.0:8080->80/tcp, [::]
:8080->80/tcp
lab_docker-logger-1     busybox                    "sh -c 'mkdir -p /da…"    logger     24 seconds ago   Up 23 seconds
PS C:\Users\Timofey\Documents\lab_Docker>
```

2.3



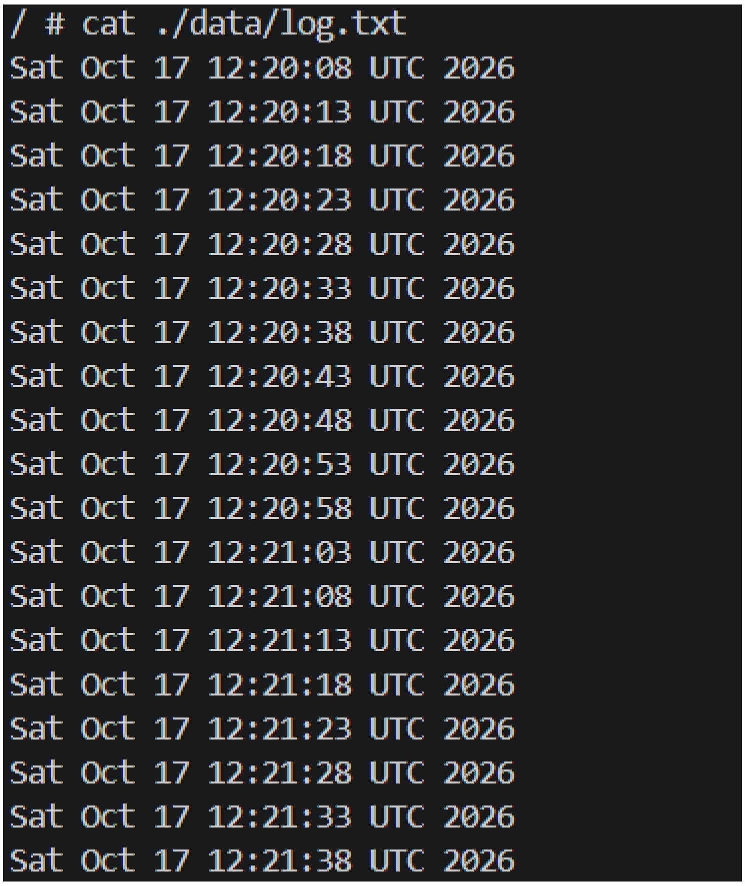
```
PS C:\Users\Timofey\Documents\lab_Docker> docker compose exec frontend sh
/ #  curl http://api
Hostname: 6b97b795d0f2
IP: 127.0.0.1
IP: ::1
IP: 172.18.0.2
RemoteAddr: 172.18.0.3:55676
GET / HTTP/1.1
Host: api
User-Agent: curl/8.22.0
Accept: */*
```

2.4 У меня не работал grep, т.к. делал на Windows, посмотрел в инете, сказали, что Select-String это аналог grep, поэтому его использовал



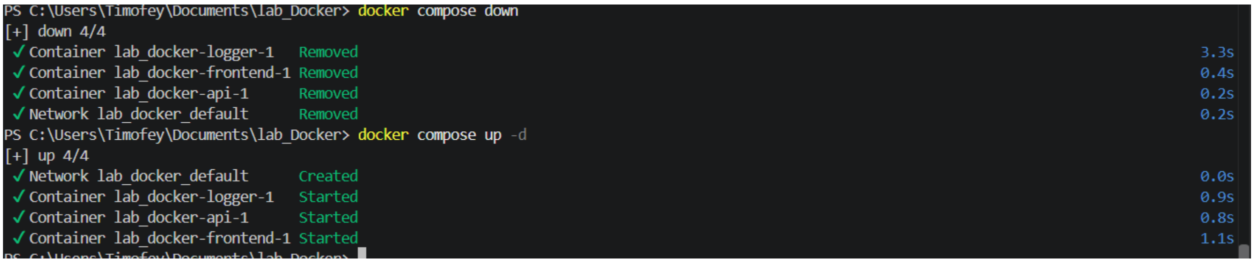
```
PS C:\Users\Timofey\Documents\lab_Docker> docker volume ls | Select-String logger

local     lab_docker_logger-data
```

2.5



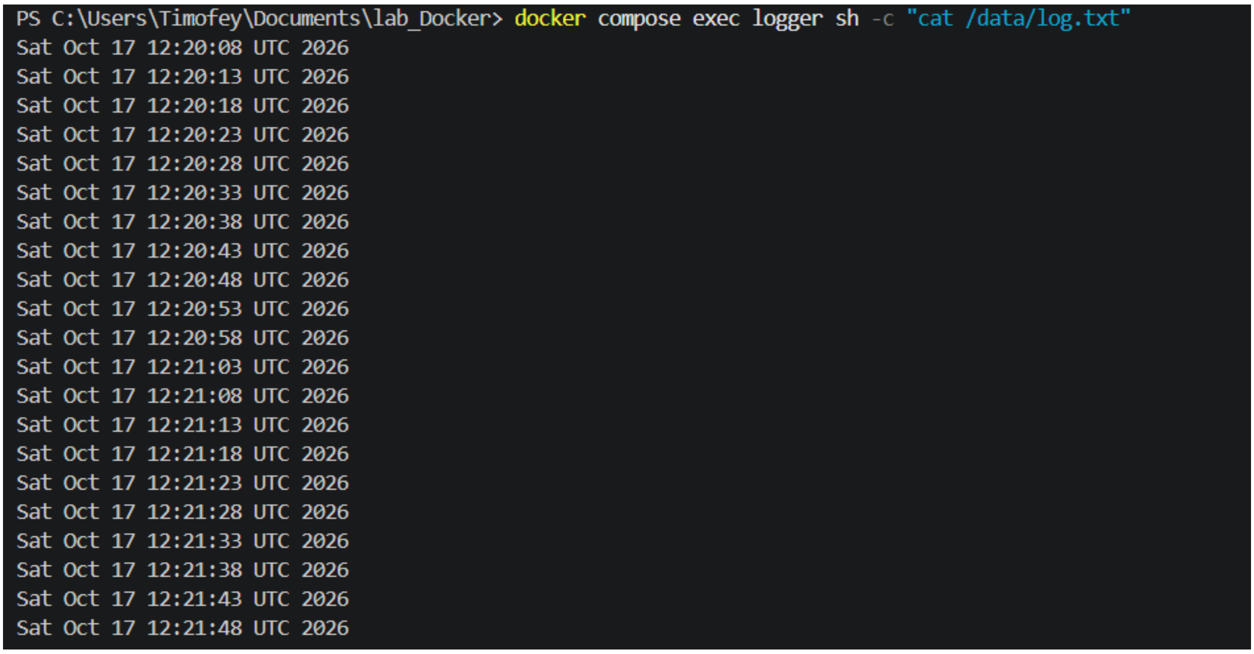
```
/ # cat ./data/log.txt
Sat Oct 17 12:20:08 UTC 2026
Sat Oct 17 12:20:13 UTC 2026
Sat Oct 17 12:20:18 UTC 2026
Sat Oct 17 12:20:23 UTC 2026
Sat Oct 17 12:20:28 UTC 2026
Sat Oct 17 12:20:33 UTC 2026
Sat Oct 17 12:20:38 UTC 2026
Sat Oct 17 12:20:43 UTC 2026
Sat Oct 17 12:20:48 UTC 2026
Sat Oct 17 12:20:53 UTC 2026
Sat Oct 17 12:20:58 UTC 2026
Sat Oct 17 12:21:03 UTC 2026
Sat Oct 17 12:21:08 UTC 2026
Sat Oct 17 12:21:13 UTC 2026
Sat Oct 17 12:21:18 UTC 2026
Sat Oct 17 12:21:23 UTC 2026
Sat Oct 17 12:21:28 UTC 2026
Sat Oct 17 12:21:33 UTC 2026
Sat Oct 17 12:21:38 UTC 2026
```

3.1



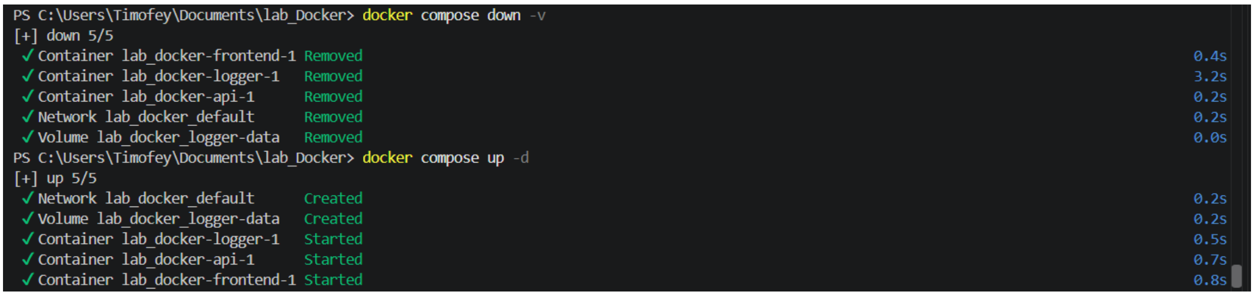
```
PS C:\Users\Timofey\Documents\lab_Docker> docker compose down
[+] down 4/4
 ✓ Container lab_docker-logger-1   Removed                                                                    3.3s
 ✓ Container lab_docker-frontend-1 Removed                                                                    0.4s
 ✓ Container lab_docker-api-1      Removed                                                                    0.2s
 ✓ Network lab_docker_default      Removed                                                                    0.2s
PS C:\Users\Timofey\Documents\lab_Docker> docker compose up -d
[+] up 4/4
 ✓ Network lab_docker_default      Created                                                                    0.0s
 ✓ Container lab_docker-logger-1   Started                                                                    0.9s
 ✓ Container lab_docker-api-1      Started                                                                    0.8s
 ✓ Container lab_docker-frontend-1 Started                                                                    1.1s
PS C:\Users\Timofey\Documents\lab_Docker>
```

3.2

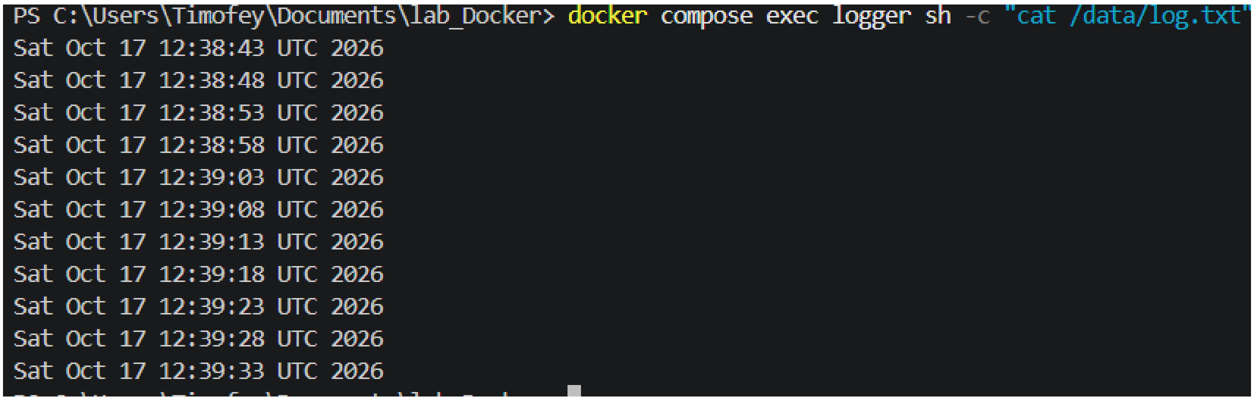


```
PS C:\Users\Timofey\Documents\lab_Docker> docker compose exec logger sh -c "cat /data/log.txt"
Sat Oct 17 12:20:08 UTC 2026
Sat Oct 17 12:20:13 UTC 2026
Sat Oct 17 12:20:18 UTC 2026
Sat Oct 17 12:20:23 UTC 2026
Sat Oct 17 12:20:28 UTC 2026
Sat Oct 17 12:20:33 UTC 2026
Sat Oct 17 12:20:38 UTC 2026
Sat Oct 17 12:20:43 UTC 2026
Sat Oct 17 12:20:48 UTC 2026
Sat Oct 17 12:20:53 UTC 2026
Sat Oct 17 12:20:58 UTC 2026
Sat Oct 17 12:21:03 UTC 2026
Sat Oct 17 12:21:08 UTC 2026
Sat Oct 17 12:21:13 UTC 2026
Sat Oct 17 12:21:18 UTC 2026
Sat Oct 17 12:21:23 UTC 2026
Sat Oct 17 12:21:28 UTC 2026
Sat Oct 17 12:21:33 UTC 2026
Sat Oct 17 12:21:38 UTC 2026
Sat Oct 17 12:21:43 UTC 2026
Sat Oct 17 12:21:48 UTC 2026
```

3.3



```
PS C:\Users\Timofey\Documents\lab_Docker> docker compose down -v
[+] down 5/5
 ✓ Container lab_docker-frontend-1 Removed                                                                                0.4s
 ✓ Container lab_docker-logger-1   Removed                                                                                3.2s
 ✓ Container lab_docker-api-1      Removed                                                                                0.2s
 ✓ Network lab_docker_default      Removed                                                                                0.2s
 ✓ Volume lab_docker_logger-data   Removed                                                                                0.0s
PS C:\Users\Timofey\Documents\lab_Docker> docker compose up -d
[+] up 5/5
 ✓ Network lab_docker_default      Created                                                                                0.2s
 ✓ Volume lab_docker_logger-data   Created                                                                                0.2s
 ✓ Container lab_docker-logger-1   Started                                                                                0.5s
 ✓ Container lab_docker-api-1      Started                                                                                0.7s
 ✓ Container lab_docker-frontend-1 Started                                                                                0.8s
```



```
PS C:\Users\Timofey\Documents\lab_Docker> docker compose exec logger sh -c "cat /data/log.txt"
Sat Oct 17 12:38:43 UTC 2026
Sat Oct 17 12:38:48 UTC 2026
Sat Oct 17 12:38:53 UTC 2026
Sat Oct 17 12:38:58 UTC 2026
Sat Oct 17 12:39:03 UTC 2026
Sat Oct 17 12:39:08 UTC 2026
Sat Oct 17 12:39:13 UTC 2026
Sat Oct 17 12:39:18 UTC 2026
Sat Oct 17 12:39:23 UTC 2026
Sat Oct 17 12:39:28 UTC 2026
Sat Oct 17 12:39:33 UTC 2026
```